

True/False

1	$\Phi(G, H)$ constant factor of $h_{\text{in}}(G)$	
	True	18
2	Remove 0.01 fraction of cycle graph	
	True	8
3	$\lambda_{\text{max}} \rightarrow 2$ implies close to bipartite	
	False	8
4	$T_{\text{mix}}(G, 0.001) = O(1)$	
	False	22
5	Transition matrix is symmetric	
	False	20
6	Diameter of Ramanujan graph	
	True	28
7	Sample pairs to find min-vertex-cut	
	False	15

8	Appending vertex r will keep $\kappa(G)$	
	True	15
9	Uniform distribution is stationary	
	True	28
10	α -lazy walk converges to stationary	
	True	29
11	Seq. of Rayleigh quotients converges	
	True	14
12	Seq. of normalized vector converges	
	False	18
13	High conductance implies robust	
	True	28
14	Robust implies high conductance	
	True	21

MCQ

1	Conductance of n -clique	
	C $(1/2 + \Theta(n))$	12
2	Sample prob. for embedding path	
	D $(1/n)$	17
3	Sample prob. for embedding clique	
	A $(1/2)$	18
4	$\sum_i \lambda_i$ for tree	
	D (0)	26
5	$\sum_i \lambda_i^2$ for tree	
	A $(2(n-1))$	22
6	$\sum_i \lambda_i^3$ for tree	
	D (0)	19